Flash Evaporation and Modelling: 101

Harsh Pandey

October 23, 2024

1 The Process

- occurs when a sudden pressure drop causes the liquid to expand beyond its saturation conditions to a meta-stable state [1].
- observed when the surrounding liquid conditions suddenly change and become lower than its saturated conditions. Due to the sudden pressure drop, the whole energy cannot be contained in the liquid as sensisble heat and the heat surplus is converted into latent heat of vaporisation [2]
- the phase change can be triggered by depressurisation under nearly adiabatic conditions. In case of hot liquids, the temperature remains almost constant till the phase change starts, but the saturation temperature drops with the pressure. As the liquid pressure reaches the saturation pressure at the corresponding temperature, boiling (formation of vapour bubbles) takes place due to the interfacial heat transfer. The bubble growth and vapour generation is controlled by the heat transfer rate at the liquid-vapor interface. The pressure undershoot (difference between saturation and pressure at boiling inception) characterises the maximal non-equilibrium between the phases - its value depending on the initial liquid temperature, depressurisation rate, available nucleation sites, and the liquid and vapour properties [3]

1.1 'Flashing Inception'

- starts when the temperature increases by a few degrees above the saturation temperature or an initirally subcooled liquid may become saturated if the pressure drop is large enough to reduce the substance pressure below the saturation pressure corresponding to its temperature [4]
- occurs when the superheat limit is reached [5]
- themodynamic limit is given by $\frac{\partial P}{\partial V}|_T = 0$ along with an equation of state that is not available really. It's the spinodal curves that show this [5]
- in practice, the metastable liquid will undergo phase change before reaching the homogenous nucleation limit [5]

1.2 Nucleation

- as a significantly high local superheat degree does not exist, the location where nucleation is initiated is determined majorly by the available nucleation sites. In most cases, the walls are a major source of nucleation sites due to roughness and manufacturing imperfectness [6]

- in a superheated liquid, fluctuations of the fluid properties on a molecular level may cause the formation of vapour nuclei. The generation of vapour nuclei can be divided into homogenous nucleation within the liquid and heterogenous nucleation at the interface between the superheated liquid and a second (typically solid) phase [1].

- in most practical circumstances, the metastable liquid will undergo phase change before it reaches the homogenous nucleation limit. In this case, nucleation occurs around pre-existing nuclei or gaseous seeds or poorly wettable cavities on solid surfaces. This is known as heterogenous nucleation [5]

- — Zone I (subcooled liquid) — Zone II (metastable superheated liquid) — Zone III (nucleation; non-equilibrium) — Zone IV (equilibrium) —

- modelling methods
  - homogenous seeding (ignores zone II and III)
  - step function (ignores zone II)
  - nucleation modelling

1.3 Bubble Growth

- at the very early stage, the nuclei are small and surface tension is dominant and restricts growth, this delay period has a marginal influence and is often ignored [7, 6]

- second stage, called inertia controlled growth [1], starts after the nuclei have grown to around twice their diameter due to the dominant inertia forces. The radius of the nucleus bubble growth is a consequence of the pressure difference between the vapour pressure in the bubble and the exterior pressure. This is hydro-dynamically controlled [6]. Bubble growth is a linear function of time [7].

- the third stage is called the thermally controlled growth mode, where the after bubble grows further and its wall temperature decreases, the increase in heat transfer between the surrounding liquid and the bubble wall causes addition of vapour to the bubble by evaporation at the interface. Bubble growth is characterised by $R \propto t^{1/2}$ [7]. Growth is restricted mainly by the rate at which heat is supplied by the liquid. However, conduction is not always the primary means of heat transfer because turbulence and relative motion between bubbles and the surrounding liquid [5]

1.4 Vapour Generation Rate

2 Modelling Multiphase Flows

Several different models and simulation approaches for various two-phase flow conditions have evolved:

- Euler-Euler: Treats the two phase flow as interpenetrating continua. The governing equations of the fluid are solved for each phase on the Eulerian grid, and the interactions of the two phases at the interface are represented through suitable exchange terms.

- Euler-Lagrange: Models treat one phase as a continuous solution on an Eulerian grid. The second phase is dispersed and treated as discrete particles in a Lagrangian framework. Here, it is assumed that the bubbles, droplets, or solids represented by the particles can be treated as rigid bodies with homogenous properties within the partile and that they are in a dilute flow. The effect of the flow field around the particles is them subsumed as appropriate forces [1]

- Multiphase Mixture: Simplified Euler-Euler approach.

Among different approaches, the homogenous equilibrium model is generally recognised as the simplest. Within this approach, the flashing flow is considered as pseudo single-phase flowing through the nozzle at the equilibrium state. The HEM approach can be applied to relatively long nozzles (that allow for a smooth phase change inside them), however, in short nozzles, this can result in large errors as the equilibrium assumption is no longer valid. To accomodate the nonequilibrium effects, empirical correction factors, slip between phases (mixture models), and two fluid models are used. The needed closure models (flow pattern details, information about bubble size distributions and bubble shapes) are not general and are selected on a case-by-case basis. Approaches to modelling flashing flows need to consider: (a) the change of phase during flashing flow and (b) the interaction between the liquid phase and the vapor phase during the vaporization process. [8]

Concering the change of phase during flashing flow, the models can be classified into two main categories:

- models accounting for the nucleation process (which show better agreement with the real behaviour but require experimental data to tune the nucleation source term)

- models neglecting nucleation and assuming constant bubble size or number density

References

[1] Jan Wilhelm Gärtner. *Modeling and simulation of flash evaporation of cryogenic liquid jets.* 2024. DOI: 10.18419/opus-14550.

[2] D. Saury, S. Harmand, and M. Siroux. "Flash Evaporation from a Water Pool: Influence of the Liquid Height and of the Depressurization Rate". In: *International Journal of Thermal Sciences* 44.10 (Oct. 2005), pp. 953–965. ISSN: 12900729. DOI: 10.1016/j.ijthermalsci.2005.03.005. URL: https://linkinghub.elsevier.com/retrieve/pii/S1290072905000931 (visited on 09/18/2024).

[3] Yixiang Liao and Dirk Lucas. "A Review on Numerical Modelling of Flashing Flow with Application to Nuclear Safety Analysis". In: *Applied Thermal Engineering* 182 (Jan. 2021), p. 116002. ISSN: 13594311. DOI: 10.1016/j.applthermaleng.2020.116002. URL: https://linkinghub.elsevier.com/retrieve/pii/S1359431120334839 (visited on 09/18/2024).

[4] Ahmad Mansour and Norbert Müller. "A Review of Flash Evaporation Phenomena and Resulting Shock Waves". In: *Experimental Thermal and Fluid Science* 107 (Oct. 2019), pp. 146–168. ISSN: 08941777. DOI: 10.1016/j.expthermflusci.2019.05.021. URL: https://linkinghub.elsevier.com/retrieve/pii/S0894177718319976 (visited on 09/18/2024).

[5] Yixiang Liao and Dirk Lucas. "Computational Modelling of Flash Boiling Flows: A Literature Survey". In: *International Journal of Heat and Mass Transfer* 111 (Aug. 2017), pp. 246–265. ISSN: 00179310. DOI: 10.1016/j.ijheatmasstransfer.2017.03.121. URL: https://linkinghub.elsevier.com/retrieve/pii/S001793101730460X (visited on 09/24/2024).

[6] Yixiang Liao et al. "Flashing Evaporation under Different Pressure Levels". In: *Nuclear Engineering and Design* 265 (Dec. 2013), pp. 801–813. ISSN: 00295493. DOI: 10.1016/j.nucengdes.2013.09.027. URL: https://linkinghub.elsevier.com/retrieve/pii/S0029549313005347 (visited on 09/28/2024).

[7] Osamu Miyatake, Itsuo Tanaka, and Noam Lior. "A Simple Universal Equation for Bubble Growth in Pure Liquids and Binary Solutions with a Nonvolatile Solute". In: *International Journal of Heat and Mass Transfer* 40.7 (May 1997), pp. 1577–1584. ISSN: 00179310. DOI: 10.1016/S0017-9310(96)00224-4. URL: https://linkinghub.elsevier.com/retrieve/pii/S0017931096002244 (visited on 10/05/2024).

[8] Quang Dang Le et al. "Computational Fluid Dynamics Modeling of Flashing Flow in Convergent-Divergent Nozzle". In: *Journal of Fluids Engineering* 140.10 (Oct. 1, 2018), p. 101102. ISSN: 0098-2202, 1528-901X. DOI: 10.1115/1.4039908. URL: https://asmedigitalcollection.asme.org/fluidsengineering/article/doi/10.1115/1.4039908/367598/Computational-Fluid-Dynamics-Modeling-of-Flashing (visited on 10/01/2024).